

HEARDS Hybrid Strategy

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Hybrid Factor Score Overview

This strategy integrates **5 core features** into a composite score, each capturing a unique market signal:

- Z-score Momentum
- Net Sentiment
- Event Volume Flag
- Realized Volatility
- Reversal Score

Feature 1: Z-Score Momentum

Objective:

Capture short-term trend strength relative to long-term historical behavior.

Construction:

- **Momentum Lookback:**
Compute % return over a lookback window (look back period determined by strategy dataframe).
- **Z-Score Normalization:**
Standardize using a **252-day rolling mean and std** to get:
$$z_momentum_t = \frac{rt - \mu_{252}}{\sigma_{252}}$$

where rt = 5-day return
- **Interpretation:**
 - Positive values → strong relative uptrend
 - Negative values → weak or mean-reverting behavior

Use Case:

Acts as a volatility-adjusted momentum signal, comparable across regimes.

Feature 2: Net Sentiment

Objective:

Capture daily market sentiment signals from trader actions.

Construction:

- **Source:** HEARDS dataset filtered for MOPS 380 prices.
- **Action Labels:**
 - **Bullish signals:** Raise, Bid (look into only trades)
 - **Bearish signals:** Lower, Offer
- **Daily Score:**
 $\text{Net Sentiment} = \#(\text{Bullish}) - \#(\text{Bearish})$
- **Implementation:**
 - Group by `updatedAtDate`, count actions.
 - Merge into main price dataframe by date.
 - Fill missing days with **neutral sentiment (0)**.

Use Case:

This feature quantifies dealer sentiment on a daily level to capture directional pressure in the physical fuel market.

Feature 3: Event Flag (Volume-Weighted Sentiment Score)

Objective:

Capture the *intensity* and *direction* of market actions using quoted trade volumes.

Construction:

- **Volume Estimation:**

For each entry:

$$\text{volume} = (\text{volume_min} + \text{volume_max}) / 2$$

- **Sentiment Weighting:**

- **Bullish Volume:** Sum of volume for Raise, Bid
- **Bearish Volume:** Sum of volume for Lower, Offer
- **Event Flag:**

$$\text{event_flag_t} = (\text{V_bullish} - \text{V_bearish}) / \text{V_total}$$

- **Handling Missing Days:**

Default to neutral (0) when no activity is observed.

Use Case:

Provides a *volume-weighted directional signal* to better reflect conviction behind market moves

Feature 4: Realized Volatility (Position Sizing Control)

Objective:

Adjust position size based on recent market volatility to stabilize risk.

Construction:

- **Volatility Estimate:**

$\text{realized_vol_t} = \text{std}(\text{daily_returns from } t-19 \text{ to } t)$

→ 20-day rolling standard deviation to measure short-term realized volatility

- **Volatility Targeting:**

Annual target: 10%

Daily target: $\text{target_vol_daily} = 0.10 / \sqrt{252} \approx 0.0063$

- **Volatility Scaler:**

$\text{vol_scaler_t} = \text{target_vol_daily} / \text{realized_vol_t}$

- Clipped at 1.0
- NaNs filled with 0

- **Position Adjustment:**

$\text{final_position_t} = \text{actual_position_t} \times \text{vol_scaler_t}$

Use Case:

Reduces risk exposure during volatile periods and scales up during calm regimes — improving Sharpe ratio and controlling drawdowns.

Feature 5: Reversal Score (Mean-Reversion Signal)

Objective:

Capture short-term overbought or oversold conditions using price deviation from recent mean.

Construction:

- **Reversal Window:**
5-day rolling mean of closing price
- **Formula:**
$$\text{reversal_score_t} = (\text{close_t} / \text{mean}(\text{close_}\{t-4 \text{ to } t\})) - 1$$
- **Handling Missing Values:**
Filled with 0 for early periods lacking sufficient data

Interpretation:

- Positive values → price above short-term average (potential overbought)
- Negative values → price below short-term average (potential oversold)

Use Case:

Acts as a contrarian signal to complement momentum and sentiment factors, helping detect short-term reversals.

Composite Score: Optimized Hybrid Factor Model

Objective:

Learn optimal weights for each signal to maximize Sharpe ratio of a binary signal-based trading strategy.

Process:

- **Feature Set:**
 - z_momentum
 - net_sentiment_daily
 - event_flag
 - realized_vol
 - reversal_score
- **Preprocessing:**
 - Drop rows with missing data
 - Standardize features with StandardScaler
- **Signal Generation Logic:**
 - Compute composite_score = $X_{\text{scaled}} @ \text{weights}$
 - Binary entry signal: top 10% of scores (threshold = 90th percentile)
 - Apply 1-day lag to avoid lookahead bias
- **Optimization:**
 - Objective: Maximize Sharpe ratio
 - Method: `scipy.optimize.minimize` using Nelder-Mead
 - Initial weights: uniform

Resulting Weights

z_momentum → 1.0036 → Strong momentum signal is rewarded
net_sentiment_daily → 0.9954 → Positive market sentiment is rewarded
event_flag → 0.9952 → Volume-weighted bullish events contribute positively
realized_vol → 1.0470 → Low realized volatility is preferred (inverse relationship)
reversal_score → 1.0037 → Mean-reversion potential is mildly positive

Interpretation:

- All five features have **positive weights**, meaning **higher values lead to stronger buy signals**.
- The weights are **roughly equal**, indicating each feature contributes meaningfully.
- realized_vol being slightly above 1 implies **the model rewards stability** — it scales down signals in volatile regimes.
- z_momentum, net_sentiment_daily, and reversal_score are near 1, reinforcing their importance in detecting directional edge.
- The combined score is thresholded at the **90th percentile** to trigger trades.

Hyperparameter Optimization: Grid Search

Objective:

Find the best combination of strategy parameters that maximize out-of-sample Sharpe ratio.

Parameter Grid:

- **Momentum Lookback:**
2 $\rightarrow$ 19 days (used in z-score momentum)
- **Signal Threshold (λ):**
0.1 $\rightarrow$ 0.9 (step = 0.1)
- **Holding Period:**
1 $\rightarrow$ 10 days (hard exit rule)

$\rightarrow$ **Total simulations:** 1,620 combinations

Hyperparameter Optimization: Grid Search

Optimization Logic:

1. For each parameter set:
 - Compute z-momentum using lookback
 - Merge sentiment & event features
 - Calculate **composite_score** using fixed weights
(from prior optimization: $[-0.0610, 0.1567, -0.0428] + \text{intercept}$)
2. Generate binary signals using threshold λ :
 - signal = 1 if score > λ
 - signal = -1 if score < $-\lambda$
3. Apply **position rules**:
 - Enter on signal, exit if signal crosses 0 or hold exceeds k_days_hold
4. **Volatility target** positions
5. Evaluate Sharpe ratio & total return

Result:

- momentum = 19
- $\lambda = 0.9$
- holding_period = 2

Backtesting

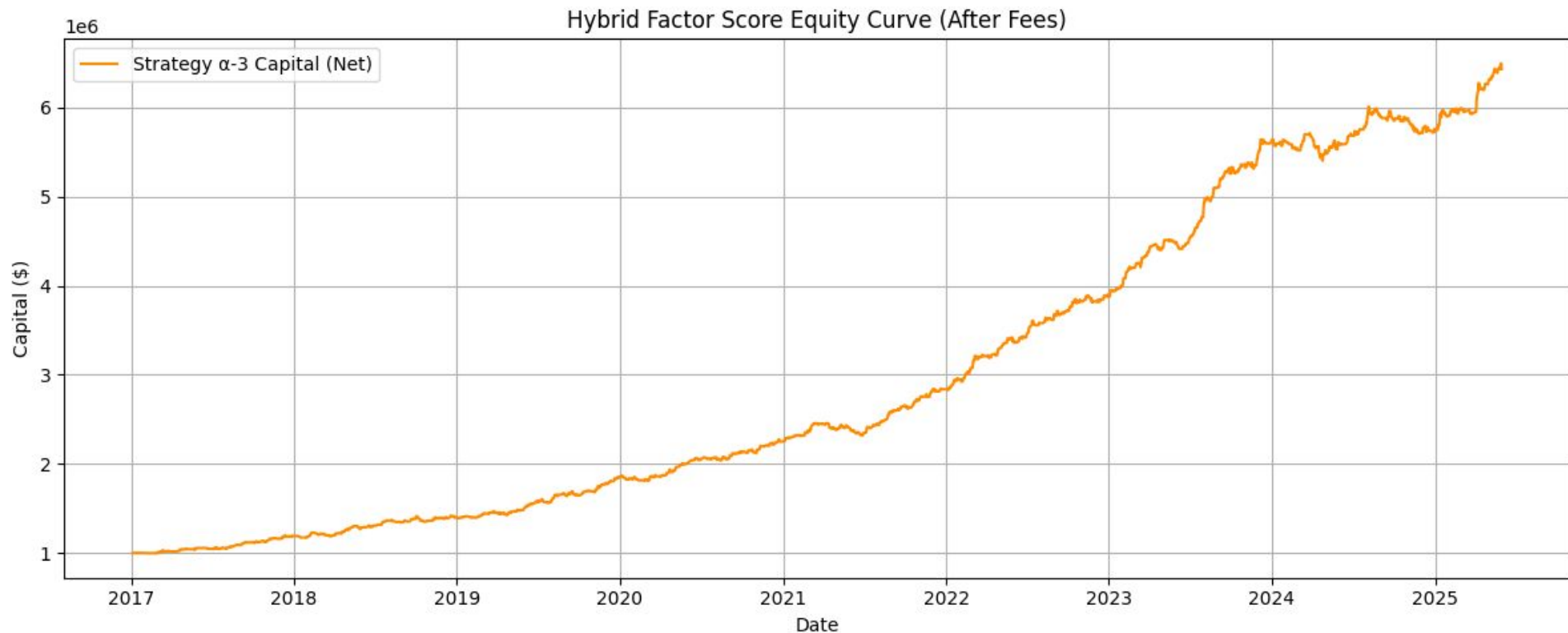
Setup:

- **Initial Capital:** \$1,000,000
- **Trading Fee:** 10 bps (0.1%) per trade
- **Positioning:** Adjusted using volatility scaling
- **Returns:** Based on $\text{final_position} \times \text{daily_return}$
- **Fees:** Deducted per position change (round trip impact)

Backtesting Results

Total Return	543.31%
CAGR	24.81%
Sharpe Ratio	3.01
Max Drawdown	-5.67%
Annual Volatility	7.50%
Win Rate	34.08%
Turnover	0.19
Avg. Hold Period	0.81 days
# of Trades	1786

Simulated Capital Growth



Drawdown Curve

